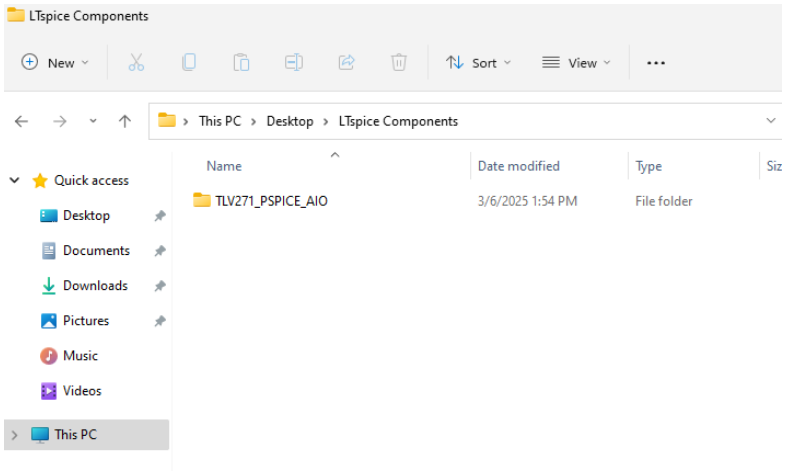
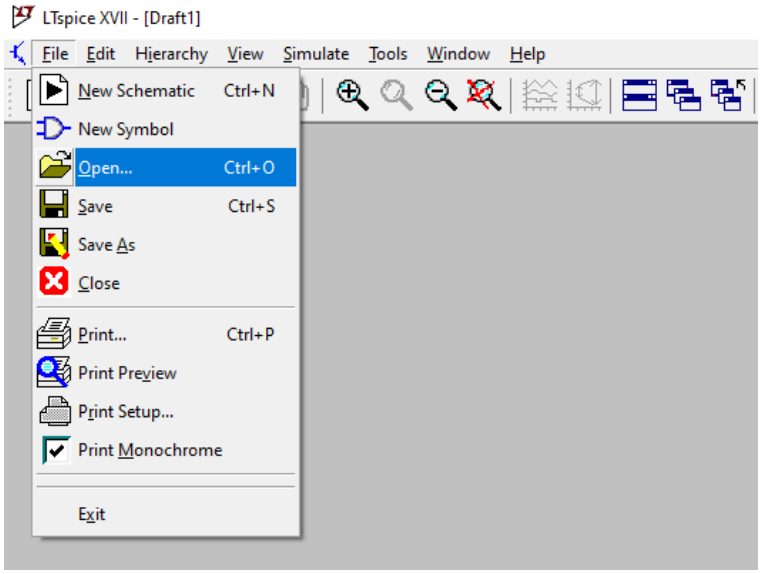
# EEL 3111C How to import components in LTspice: Windows Edition

If you are struggling to follow along with the video instructions from Lab 5 or 8, this document should hopefully help.

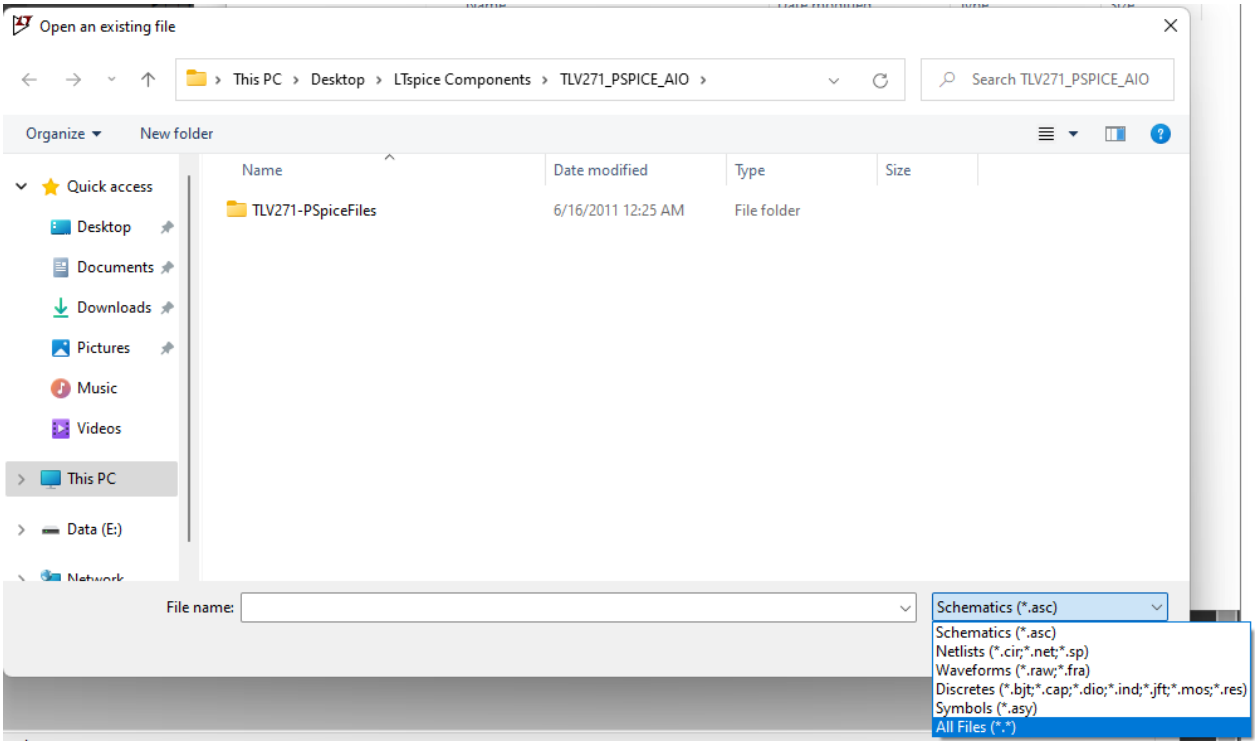
1. Download the zip file from canvas. For the op amp in lab 5: “slom249.zip”. For the comparator in lab 8: “LM393sljc016.zip”. These zip files should automatically be saved to your Downloads folder. Extract the contents of the zip file by right clicking, then save these files to a directory you have access to. For simplicity, you can create a folder on your desktop and save all component files there. The image below is for the op amp from lab 6.



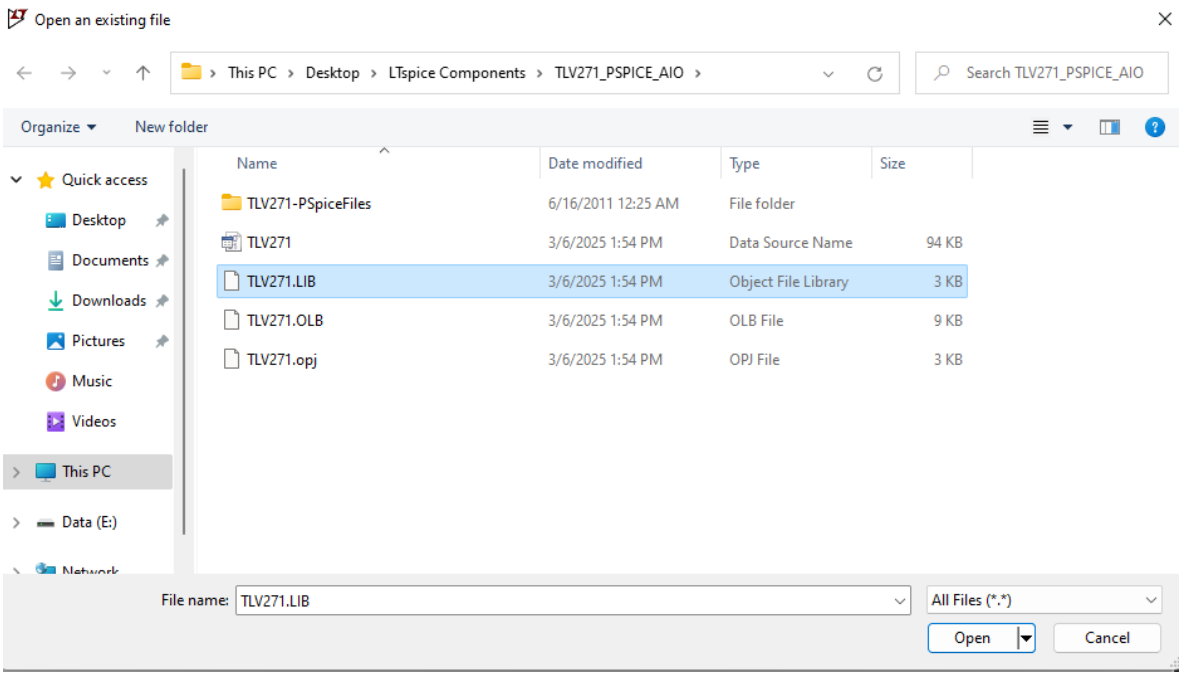

2. Open up LTspice and open a new schematic. From there you can click File->Open or press Ctrl+O.




3. Navigate to the extracted file, and look for the “.LIB” file. You might have to change the file settings to “All Files” as shown below. For the comparator in lab 8, you open the LM393.5_1 file.



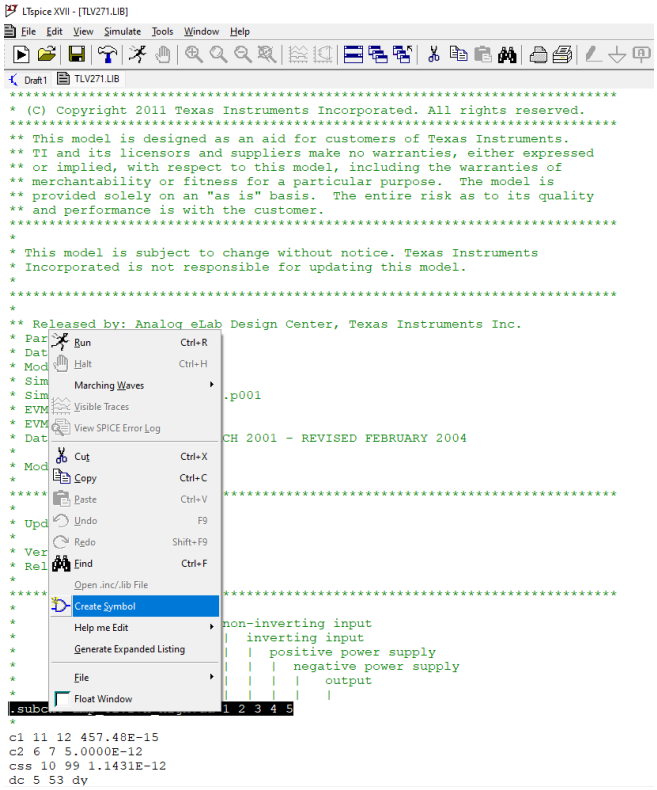

4. Find the ".LIB" file. For the op amp its "TLV271.LIB". Once again, for lab 8 you open the "LM393.5_1" file, there's no ".LIB" file for the comparator. If this doesn't work, drag and drop the file into LTspice.



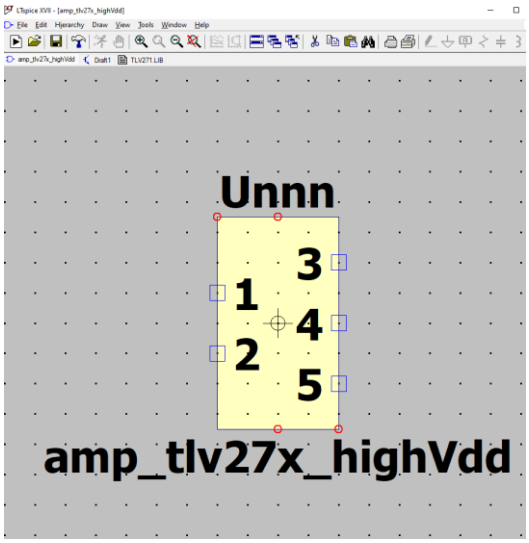
5. Once you open the file, you should see the image below. This tells you all the info you need about the component, including the connections, model parameters, and datasheet name if needed.



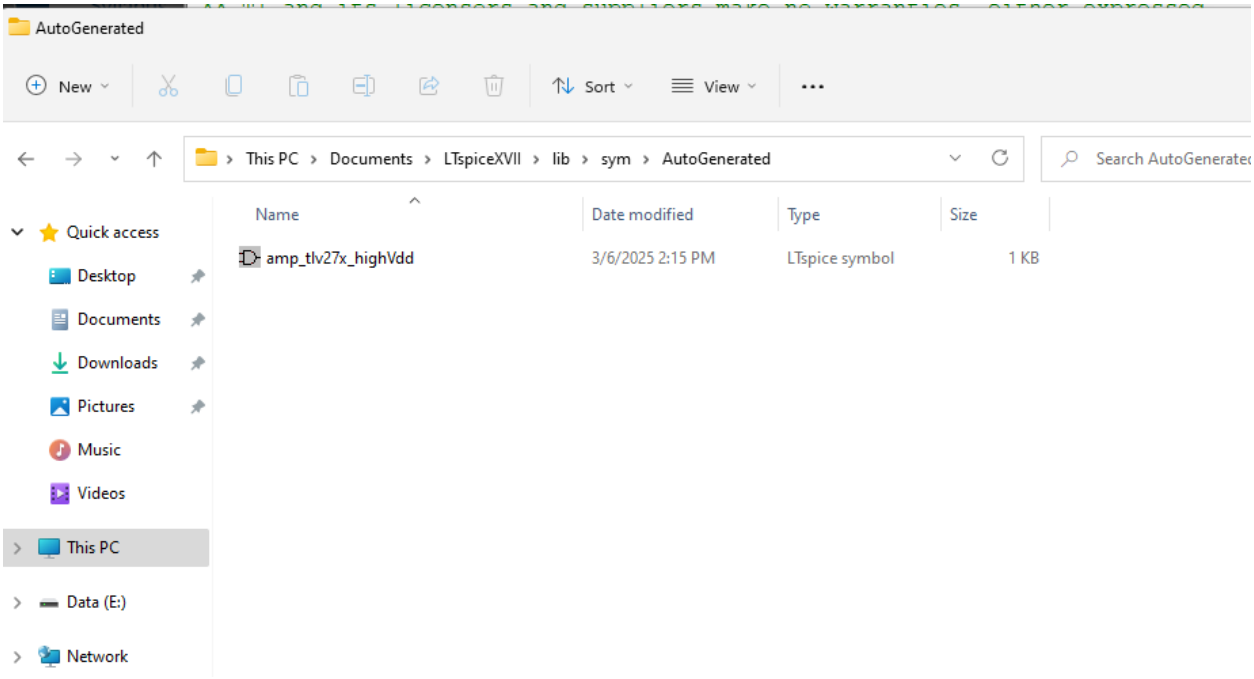

6. To create the new symbol, highlight the blue text ".subckt amp_tlv27x_highVdd 1 2 3 4 5" and right click to press Create Symbol.



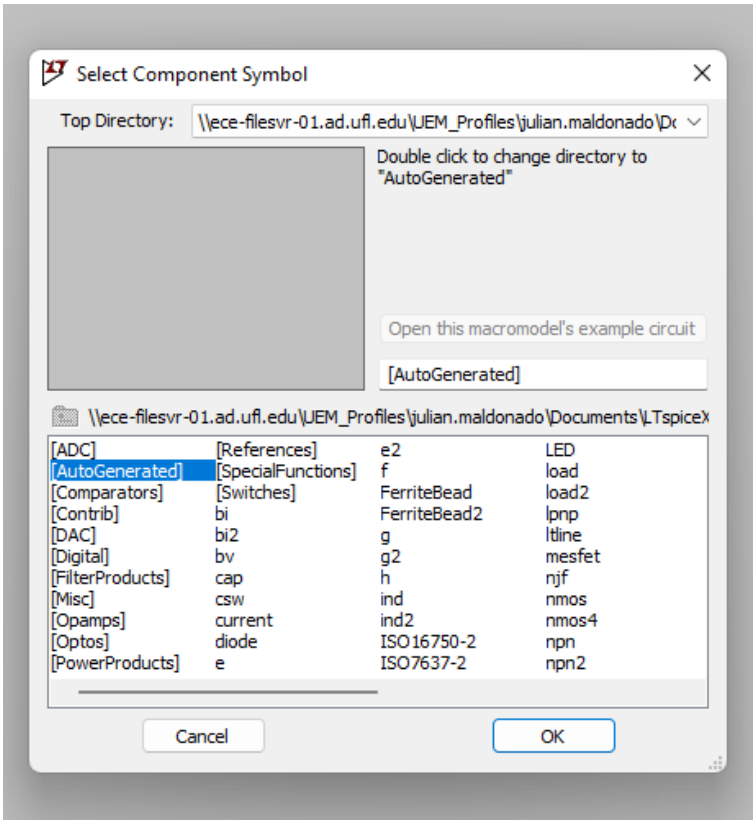

7. A window might pop up asking if you want to automatically create a new component, just press Yes. The following should now appear on your screen. That's your new symbol! If you wish to change the shape, refer to the video linked in the lab manual but it might cause you errors. In case you mess up and can't figure out what's wrong, just delete the files and restart again and use the default shape.

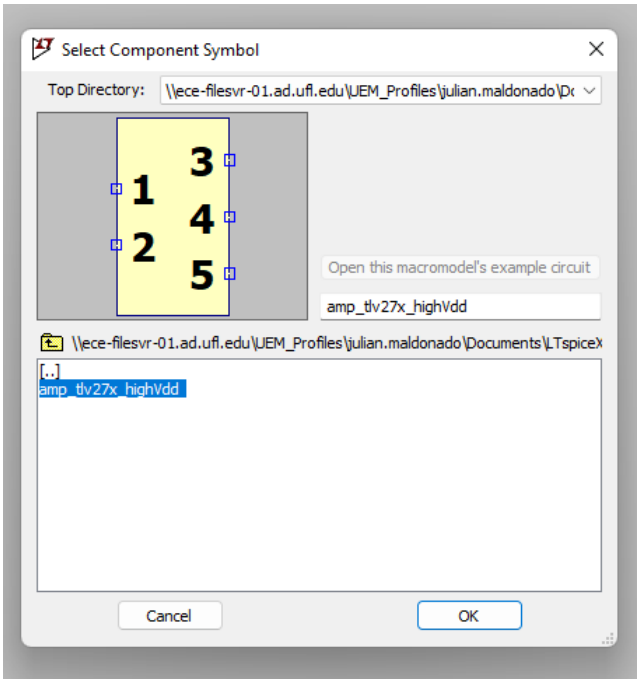



8. The symbol file should automatically save to "/Documents/LTspiceXVII/lib/sym/AutoGenerated" If it's not there, you must place it there yourself. If you want, you can also go that directory in your file explorer and create a new folder to place your downloaded components.




9. Go back to your schematic and start building your circuit. To access your new component, right click the screen->Draft->Component and then open the AutoGenerated folder. Your component should be there.






10. Press ok and place your component down. Continue to build your circuit as normal. You did it! Below is an example of an inverting op amp circuit. For the comparator, all steps are the same after you open the “LM393.5_1” file.

